The Power User Trap℠: When AI Adoption Becomes a Retention Risk

Organizations are deploying AI tools to improve efficiency and scale productivity. The technology works because someone makes it work. That burden does not distribute evenly.

In most organizations, a small group of employees becomes the go-to resource for AI implementation. They learn the tools first, troubleshoot failures, and train colleagues. These employees are usually high performers with deep institutional knowledge who must function in a continuous state of doubt. They hold judgment: Is the output reliable? Should I verify this manually? Where is the hallucination?

They are the "Power Users" and when they leave, organizations lose both the expertise and the implementation capacity. The exit is coded as personal.

This is the Power User Trap℠: employees operating in continuous doubt, constantly verifying and validating AI outputs without organizational recognition of the cognitive burden. It represents one mechanism through which Invisible Attrition℠ occurs, where leadership capacity erodes before retention metrics detect risk. HR systems record the departure but not the cause.

The Unmeasured Cognitive Cost

Research from Boston Consulting Group confirms what many organizations are experiencing but not measuring. In a March 2026 study in Harvard Business Review of 1,488 full-time U.S. workers, researchers found that workers providing high degrees of AI oversight expended 14% more mental effort and experienced 12% more mental fatigue than those with low oversight requirements. The study identified what researchers termed "mental fatigue from excessive oversight of AI tools beyond one's cognitive capacity."

The cognitive strain carries measurable business costs. Workers experiencing mental fatigue from intensive AI oversight reported making major errors 39% more frequently. Decision fatigue increased by 33%. Intent to quit rose from 25% to 34%.

For HR leaders, this represents a retention risk that current measurement systems are not designed to capture.

Where the Risk Concentrates

The BCG research found significant variation by function. Marketing roles reported the highest mental fatigue rate at 26%, followed by HR at 19%, operations at 18%, engineering at 18%, and finance at 17%. These are knowledge work roles where high performers adopt AI tools quickly, train others, and maintain productivity while managing implementation burden. The work rarely appears in job descriptions, performance reviews, or workload assessments.

One senior engineering manager in the study said: "I was working harder to manage the tools than to actually solve the problem."

This is the operational signature of the Power User Trap℠. The employee appears productive in output metrics while absorbing unsustainable cognitive load privately. HR sees the exit after the retention opportunity has passed.

What Can HR Measure?

Preventing this may require a shift from monitoring AI tool adoption rates to monitoring AI implementation burden. Organizations are being asked not only what AI decisions were made, but how those decisions were made, applied, and documented. When implementation burden goes unmeasured, organizations deploy AI without understanding who is carrying the operational cost.

First, limit simultaneous AI tool use. The BCG study found that productivity peaked at three tools used simultaneously, then declined. Organizations measuring token consumption or lines of AI-generated code as performance metrics may inadvertently incentivize cognitive overload.

Second, clarify workload expectations explicitly. When employees felt their organization would expect them to accomplish more work due to AI, mental fatigue scores were 12% higher.

Third, track oversight distribution directly. Who is being asked to support AI tool adoption beyond their formal role? Are those individuals receiving workload adjustments or compensation changes? How many hours per week are high performers spending on AI-related support work?

From Deployment to Accountability

AI tool deployment is often treated as a technology decision. The governance implications are workforce sustainability and leadership continuity.

When organizations adopt AI without measuring how implementation burden is distributed, they create retention risk they cannot see. Power users absorb the cost privately. Workload remains undocumented. Exit data only shows voluntary departure.

For HR leaders facing increasing scrutiny over workplace decisions, this represents a measurable gap. When AI influences a workplace decision, accountability still sits with HR.

The BCG study also found that 19% of HR professionals using AI reported experiencing elevated cognitive strain from AI oversight. The function responsible for managing workforce health may itself be caught in the Power User Trap℠.

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